Personalizing POI Descriptions with LLMs: A User Study of Perceived Information Quality

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Abstract

Many Points of Interest (POI) descriptions remain generic and static, presenting identical content to all users regardless of interests or travel preferences. This can limit perceived relevance and may reduce engagement with travel platforms. This study examines whether large language models (LLMs) can personalize POI descriptions to improve perceived information quality. In a within-subjects study, 51 participants evaluated original and LLM-adapted descriptions across 10 POIs. Paired statistical analyses reveal that AI-adapted descriptions achieved significantly higher ratings for both clarity ($p = .002$) and trustworthiness ($p < .001$), while perceived significance remained comparable between conditions ($p = .154$). Optional post-study responses indicate high comfort with AI-generated travel content ($M = 3.94$, $n = 17$). These findings suggest that LLM-based personalization can enhance perceived information quality in tourism micro-content without diminishing credibility or engagement.

CCS Concepts

• **Human-centered computing → User centered design**; *User studies.*

Keywords

poi adaptation, LLM-generated content, ai-driven descriptions, content personalization

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1 Introduction

Travel planning increasingly happens through digital platforms that help users evaluate accommodations, attractions, and other Points of Interest (POIs). Recommender systems have therefore become central in tourism applications, where personalization is used to cope with information overload and heterogeneous user needs [1, 7]. In the POI domain specifically, prior work has largely emphasized predicting which places users may prefer (e.g., next-POI or ranked POI recommendation) and how to incorporate richer signals such as context and heterogeneous data [5, 10]. Recent surveys further argue that large language models (LLMs) are becoming an enabling component across the recommendation pipeline, including user interaction and explanation generation [8, 9, 13].

However, while ranking and recommendation determine *which* POIs are shown, users still rely on POI *descriptions* to decide *why* a place is relevant and whether it is credible and worth visiting. In many systems, these descriptions remain static and generic, presenting the same framing to all readers despite differences in interests, travel styles, and constraints. This "one-size-fits-all" presentation can under-emphasize what matters most to a given user (e.g., historical context versus family suitability), potentially limiting perceived clarity and usefulness. Generative AI offers a way to adapt the *emphasis* of descriptions at scale, complementing POI recommendation and supporting more user-centered travel decision-making [4, 12].

At the same time, deploying LLM-generated travel content raises practical concerns around information quality and trust, especially in domains where credibility is critical. Tourism-focused GenAI research has begun to examine opportunities and risks (e.g., reliability, domain adaptation, and user acceptance), and approaches such as retrieval augmentation have been proposed to mitigate factual errors [3, 4, 6]. Motivated by this gap, this study examines whether LLM-based personalization of POI descriptions can improve perceived information quality without harming trust or engagement. A prototype was developed that presents users with original (manual) descriptions and LLM-adapted variants conditioned on stated preferences, and both versions were evaluated in a within-subjects study across multiple POIs. The contributions of this study are (1) an empirical comparison of manual versus LLM-adapted POI descriptions using significance, trustworthiness, and clarity as evaluation dimensions, and (2) evidence about user acceptance of adaptive description personalization in a tourism context.

2 Related work

Prior work on tourism and location-based personalization has largely focused on point-of-interest (POI) recommendation, where systems predict or rank places that match a user profile and context. Surveys of travel and POI recommenders highlight the importance of heterogeneous signals such as mobility patterns, time, geography, and user attributes, as well as evaluation challenges that arise from contextual variability and data sparsity [7, 10]. Recent work in the UMAP community also emphasizes that user preferences can differ before and after a visit, which complicates modeling and suggests that preference signals and content expectations are not static [2].

With the rise of large language models (LLMs), research has expanded to interactive and generative paradigms for personalization, including LLM-based pipelines for POI recommendation and user profile generation [5, 8, 9, 11]. Tourism research has also begun evaluating generative AI for content creation and trip planning [3, 4, 12]. However, these works often study destination promotion or end-to-end planning rather than the micro-content users read when deciding whether a POI is relevant. This study addresses this

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gap by adapting POI descriptions to user interest profiles while keeping core facts stable, measuring perceptions of significance, trustworthiness, and clarity.

3 Methodology

This study examines how users perceive LLM-adapted POI descriptions in comparison to original, manually written descriptions. It focuses on perceived information quality, specifically clarity, trustworthiness, and relevance, rather than on recommendation accuracy or task performance. We conducted a user-centered comparative study in which participants evaluated paired descriptions of the same POI. Each pair included an original description sourced from existing online material and an LLM-adapted version personalized using the participant's stated preferences. Participants were not informed which description was AI-generated, ensuring that evaluations reflected perceived qualities rather than source awareness.

3.1 Study Design

A within-subjects experimental design was employed in which each participant evaluated both original and AI-personalized descriptions for multiple POIs. This design enabled direct comparison between description types while controlling for individual differences in preferences and expectations. The study presented participants with 10 POIs spanning five categories (two per category): accommodation (Hotel Hassler Roma, The Savoy), natural attractions (Jungfraujoch, Neuschwanstein Castle), entertainment (Cologne Carnival, Cannstatter Volksfest), cultural and historical sites (Colosseum, Louvre Museum), and commercial destinations (Via del Corso, Café Central Vienna). POIs were selected to include both internationally recognized locations (e.g., Colosseum, Louvre) and regionally significant destinations (e.g., Cannstatter Volksfest), providing variety in familiarity levels.

3.2 Participants

A total of 51 participants were recruited through personal networks via email and direct messaging. Participants were required to be at least 18 years old and possess basic English proficiency to ensure effective engagement with POI descriptions. No specific travel experience was mandated, as the study aimed to assess AI-generated descriptions across diverse user segments.

Participants ranged from 22 to 49 years old ($\mu = 30.94$, $\sigma = 6.74$), with 67% male and 33% female. The sample was predominantly Indian (n = 28) and German (n = 21), with additional participants from China (n = 1) and Iran (n = 1). Professionally, participants included students (n = 22), engineers (n = 16), academics (n = 3), and other professions (n = 10). Most reported intermediate (n = 21) or experienced (n = 19) travel levels. The sample's geographic concentration reflects convenience sampling through personal networks rather than stratified recruitment; accordingly, findings should be interpreted as applicable to technology-familiar, educated populations with similar demographics.

3.3 LLM-Based Personalization of POI Descriptions

To operationalize personalization in the study, we employed a lightweight and explicit user profile derived from participants' self-reported attributes and preferences. The profile was not intended as a predictive user model, but rather as a structured representation of personal context used to guide content adaptation. Personalization focused on adjusting emphasis and framing within POI descriptions while preserving their original meaning and factual content.

For each POI, two description versions were prepared. The original description was sourced from official tourism websites and travel platforms, representing standard, non-personalized content. The AI-adapted version was generated using gpt-4o-2024-11-20[1] with the default temperature setting. The model received a system prompt instructing it to act as an expert travel writer, emphasizing unique aspects, cultural significance, and visitor experience while adapting content to the user's situation. The user prompt included ten personalization dimensions: age, gender, marital status, whether the user has children, interests, travel experience level, education, profession, hobbies, and preferred travel style. The original POI title and description were provided, and the model was instructed to maintain factual accuracy and length, while emphasizing aspects most relevant to the user's profile. To verify factual preservation, a subset of generated descriptions was manually reviewed during initial testing to confirm that no factual claims were altered or fabricated during personalization. For example, a museum description for a participant interested in history would emphasize historical significance and artifacts, while the same museum described for a family traveler would highlight interactive exhibits and educational programs.

3.4 Evaluation Interface

Participants viewed descriptions through a web-based interface that displayed two versions side by side, labeled as Version A and Version B (see Figure 1). The interface did not indicate which version was AI-generated. Each POI was accompanied by a representative image to provide visual context. The presentation order of AI and manual descriptions was randomized across POIs to prevent order bias. Participants could not proceed to the next POI without completing evaluations for the current one, ensuring complete data collection.

3.5 Information Quality Measures

Participants assessed each POI description along three perceptual dimensions using 5-point scales:

- **Significance**: "Does the description effectively communicate the significance and offerings of the place?" (5-point agreement scale: 1 = Strongly Disagree, 5 = Strongly Agree)
- **Trustworthiness**: "How trustworthy does this description appear to be?" (5-point intensity scale: 1 = Not at all, 5 = Extremely)
- **Clarity**: "How clear and complete is the information provided?" (5-point clarity scale: 1 = Very Unclear, 5 = Very Clear)

[1] https://platform.openai.com/docs/models/gpt-4o

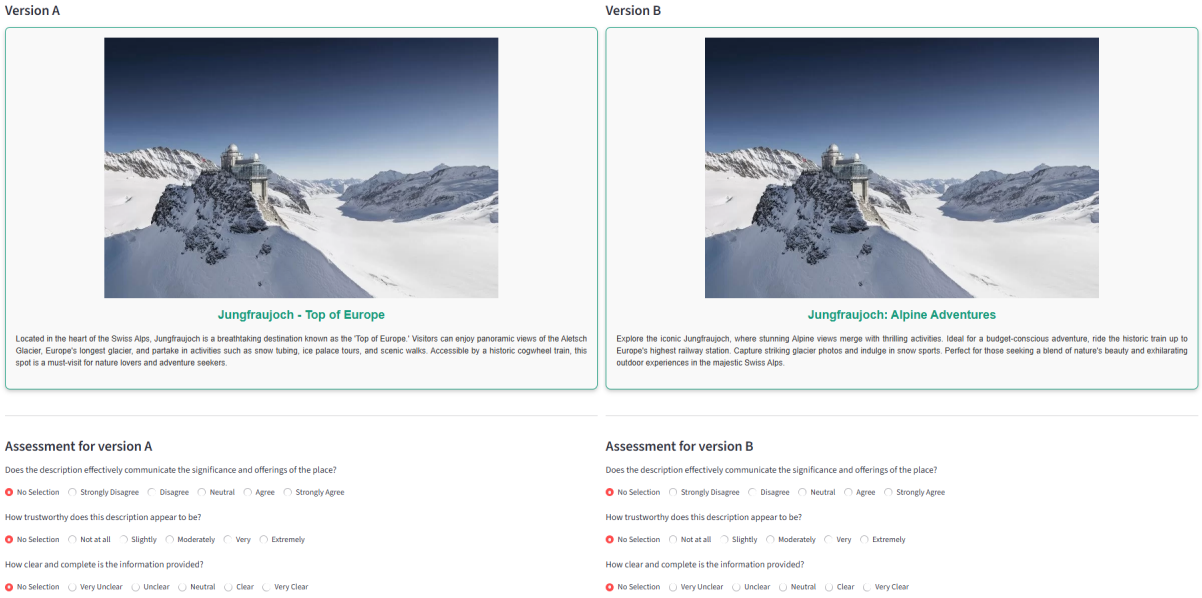

Figure 1: Evaluation interface showing two description versions (Version A and Version B) presented side by side for participant comparison

After rating both descriptions individually, participants completed comparative judgments indicating which description was more engaging, informative, appealing, and more likely to encourage a visit. Participants also reported their prior familiarity with each POI (visited, heard of, or unfamiliar).

3.6 Procedure

The study was conducted in three phases using a secure web platform. First, participants reviewed an informed consent form explaining the study purpose, voluntary participation, and data confidentiality. After providing consent, they completed a demographic and preference survey collecting age, gender, nationality, profession, current city, hobbies, preferred travel styles, and tourism interests. This information formed the basis for personalizing AI-generated descriptions. The interface required a minimum number of preference selections to ensure sufficiently expressive profiles. Based on this information, POI descriptions were automatically generated using a large language model in subsequent pages of the study.

Participants then proceeded to the main evaluation phase, where they were presented with POI descriptions adapted using their profile information alongside original descriptions. The adapted descriptions were generated dynamically after the profiling step and before the evaluation, without further user interaction. Second, participants evaluated 10 POIs sequentially by reading both description versions and rating them on core metrics and familiarity. This evaluation phase lasted approximately 20 to 30 minutes. Finally, participants provided overall feedback on their experience and their comfort level with AI-generated travel content. All data were collected anonymously to ensure privacy. The study was conducted in accordance with institutional ethical guidelines, with participants providing informed consent before participation.

4 Results

This section presents findings from the user study comparing LLM-adapted POI descriptions against manual baselines. Across 51 participants evaluating 10 POIs each, the study collected 510 paired ratings per metric (manual versus AI). Paired statistical tests were employed to account for the within-subjects design, where each participant evaluated both description types for the same POI. Table 1 summarizes mean ratings and statistical outcomes, while Figure 2 visualizes mean differences with 95% confidence intervals.

Perceived Significance. Both description types received favorable ratings, and the difference was not statistically significant ($p = .154$, $d = 0.08$). This suggests that LLM-based adaptation maintained perceived significance without diminishing users' understanding of what each location offers. The result suggests that personalization did not appear to distort or weaken the core message about a POI's importance and unique characteristics.

Trustworthiness. AI-generated descriptions received higher trustworthiness ratings ($p < .001$, $d = 0.19$). This result is interesting given common concerns about AI-generated content credibility [4]. One possible explanation is that personalization may have enhanced perceived relevance by highlighting aspects aligned with participants' stated interests, potentially contributing to perceptions of accuracy and reliability.

Clarity and Completeness. AI-generated descriptions were rated significantly higher on clarity ($p = .002$, $d = 0.17$). This finding aligns with the personalization goal: descriptions that emphasize details matching user interests may be easier to read and understand. By focusing on relevant information, personalized descriptions may reduce the effort required to extract useful content, potentially helping users process the same facts more efficiently.

Metric	Manual	AI	t	p	d
Significance	3.73 ± 0.78	3.79 ± 0.83	−1.43	.154	0.08
Trustworthiness	3.56 ± 0.85	3.71 ± 0.82	−3.43	**<.001**	0.19
Clarity	3.70 ± 0.80	3.84 ± 0.80	−3.14	**.002**	0.17

Table 1: Paired comparison of manual vs. AI descriptions (N=510, df=509). Values show $\mu \pm \sigma$ on 5-point scales. Negative t-values indicate AI rated higher. Cohen's d reports effect size; $p < .05$ bolded.

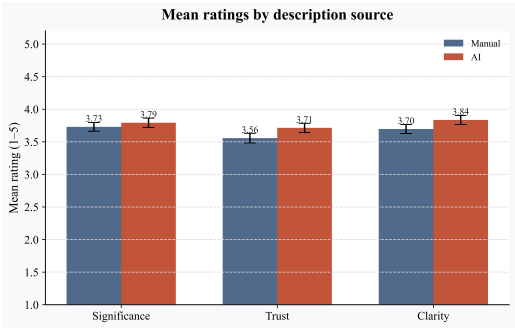

Figure 2: Mean ratings comparison between manual and AI-generated descriptions. Error bars represent 95% confidence intervals.

4.1 Comparative Preferences

Beyond individual ratings, participants completed comparative preference questions, selecting which description they found more engaging. Responses were nearly evenly distributed: 221 selections (43.3%) favored AI-generated descriptions, 220 selections (43.1%) favored manual descriptions, and 69 selections (13.5%) indicated both versions were equally engaging. This parity suggests that AI-adapted descriptions achieved comparable engagement to manual descriptions in our sample. Participants did not appear to perceive the AI-adapted content as less interesting or appealing, suggesting that personalization may improve clarity and trustworthiness without compromising subjective engagement.

4.2 Overall Study Feedback

A subset of 17 participants completed optional feedback ratings (5-point scale). Those who provided feedback were demographically similar to those who did not. Overall experience received positive ratings ($\mu = 3.88$, $\sigma = 0.70$). Participants expressed moderate support for automatic POI adaptation ($\mu = 3.75$, $\sigma = 0.93$) and high comfort with AI-generated travel content ($\mu = 3.94$, $\sigma = 0.83$), suggesting receptiveness to AI-generated descriptions when demonstrating quality and relevance.

5 Discussion

This study examined whether LLM-based personalization of POI descriptions changes how people judge the quality of travel information. Paired statistical analyses revealed that AI-adapted descriptions received significantly higher ratings for both clarity and trustworthiness, while perceived significance remained comparable between conditions. One possible explanation is increased perceived relevance: when descriptions foreground details aligned with a reader's interests, the same factual content may appear easier to process and more credible. However, other mechanisms such as writing style differences or content organization may also contribute to these effects.

Although the differences in mean ratings were modest in absolute terms, the consistency of improvement across both clarity and trustworthiness, combined with comparable engagement ratings, suggests that personalization may provide benefits without introducing obvious drawbacks.

From a practical perspective, these preliminary findings suggest potential for using LLMs to adjust emphasis and framing in POI micro-content using lightweight preference signals. This approach may help scale personalization without extensive manual editing. The improvement in trustworthiness is particularly noteworthy, as it addresses a common concern about AI-generated content [4]. Platforms implementing such systems may consider transparency and disclosure as design choices. While this study did not manipulate transparency (participants were unaware which descriptions were AI-generated), future work should investigate how disclosure affects perceived quality and trust.

5.1 Limitations and Future Work

Several limitations should be acknowledged. The controlled study setting may not fully reflect real-world travel planning, and we measured participants' perceptions rather than actual decisions (e.g., whether they would book a place). The sample composition (67% male, two nationalities, convenience sampling) limits generalizability, and results are based on a single LLM with default settings. Effect sizes, though significant, were small. Future work should expand participant diversity, conduct field studies tracking actual behavior, and test finer personalization factors (trip purpose, companions, budget). Investigating transparency about AI involvement and balancing personalized relevance with serendipitous discovery remain open questions.

6 Conclusion

This study provides preliminary evidence that large language models can personalize POI descriptions in ways that may improve how users judge travel information. In our sample, AI-adapted descriptions received higher ratings for both clarity and trustworthiness, while perceived significance remained comparable. Optional post-study feedback suggests that participants were generally comfortable reading AI-generated descriptions in this setting.

These findings suggest that personalization may support travel decision-making by foregrounding details that match user interests while preserving core facts about a place. For travel platforms, LLM-based rewriting may offer a path toward scaling tailored micro-content without extensive manual editing. More broadly, these initial results point toward the possibility that description framing could shape perceived information quality alongside which places are recommended.

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